SHRIKANT SHARMA

774-438-9698 | shrikant.sharma@myyahoo.com | shrikant-sharma.github.io | Berlin, NH | Open to Relocation
linkedin.com/in/shrikant-sharma | github.com/Shrikant-Sharma | public.tableau.com/app/profile/shrikant.sharma

Data Scientist | Applied AI, LLM Evaluation & Machine Learning | Python, SQL, LangGraph, AWS

Data Scientist with ~8 years in regulated pharma and financial services, building the pipelines, statistical models and evaluation methods that compliance and risk leaders act on: a federal submission pipeline (~509K records, rejections under 0.5% across 4 cycles), a risk index Directors used to allocate resources, and an A/B testing framework for credit-risk experiments. Recent depth in LLM applications and their evaluation through three live public projects, with model selection, calibration and causal inference shown end to end.

SKILLS

Machine Learning & AI: LLM Evaluation (LLM-as-a-judge, faithfulness, test question sets, error analysis), Agentic AI, RAG & Vector Search, Causal Inference (PSM, G-Computation), Experiment Design & A/B Testing, MLOps, Feature Engineering, Guardrails, Prompt Engineering, Model Governance, NLP, Gradient Boosting, Deep Learning, Anomaly Detection, Segmentation, Time-to-Event Forecasting, Calibration, Explainability (SHAP)

Languages & Libraries: Python (Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, Hugging Face Sentence Transformers, FAISS, SHAP, LangGraph, DoWhy, lifelines, PySpark, Matplotlib), SQL, R

Cloud & Data Platforms: AWS (S3, Lambda, ECR), Databricks, Snowflake

Tools & Platforms: MLflow, Docker, Git, FastAPI, Apache Spark, Model Context Protocol (MCP), Streamlit, Tableau

EXPERIENCE

Amgen | Data Scientist **Sept 2022 – May 2026**

- Rebuilt the half-built pipelines behind Amgen's federal CMS Open Payments submission into validated Databricks pipelines on AWS (~509K records, ~$630M): SQL validation gates reconciled counts at every stage and mapped every government error code, holding rejections under 0.5% across 4 annual cycles and cutting cycle time from 2 weeks to 4 days; built the year-over-year Tableau dashboards compliance used to review each cycle.
- Designed the statistical risk index (peer-group z-scores with tuned thresholds across 20+ indicators) that Directors and Executive Directors used to allocate compliance resources across 65+ ex-US countries and 40+ US products; chose the indicators with compliance leads in 3 regions, combining system data with their own assessments; caught score drift between annual cycles by comparing distributions, then rebuilt and revalidated the index.
- Proposed and built a document classification prototype (TF-IDF and Random Forest; Python, Scikit-learn) for 4 types of compliance documents (audit reports, investigation summaries, policy and SOP documents, monitoring reports), delivered as an on-request Databricks job for the compliance team and validated on held-out documents.
- Built statistical anomaly detection for transaction monitoring across 5 payment processes and 50K+ records in the US, ELMAC and JAPAC regions: peer-group z-scores with tuned thresholds and flag-rate stability tracked across runs to catch drift.
- Contributed to 2 company-wide GenAI hackathons, building proof-of-concept LLM applications for audit-finding summarization and question answering over regulatory documents and checking outputs against the source text.
- Mentored 3 junior data scientists on Python automation, modeling practice and reproducibility; wrote the analytics workflow documentation the team adopted as its reference; delivered 3 concurrent projects on schedule.

Nugget.ai | Data Scientist Intern **Jan 2022 – May 2022**

- Trained models in Python (Scikit-learn) to predict candidate-role fit from behavioral assessment data, engineering 20+ features spanning structural, linguistic, behavioral, and content-based signals. Compared Logistic Regression, Random Forest, and Gradient Boosting with cross-validation, improving matching accuracy 15% over the platform baseline.

Atos Syntel | Data Scientist | *Client: American Express* **Feb 2018 – Dec 2019**

- Built the production risk-scoring pipelines in Python and SQL over 10M+ transactions, deriving behavioral features (spending velocity, geographic patterns, customer segments) that fed the fraud and credit risk teams' decisioning models and the weekly review.
- Owned the A/B testing framework for credit-risk strategy experiments in Python, SQL and Git: hashed customer-ID assignment, one primary metric with guardrails, power-based sample sizing, Bonferroni and Benjamini-Hochberg corrections, and sample-ratio and peeking checks across 12+ cycles a year.
- Designed the Tableau dashboards that consolidated 5+ source systems into the single executive view used in the weekly credit risk review, supporting mitigation decisions totaling over $2M.

Atos Syntel | Data Analyst | *Client: American Express* **Mar 2016 – Jan 2018**

- Built the SQL pipelines and validation checks behind monthly regulatory reporting on financial transaction data, and automated the recurring reports in SQL and Python so the team's time moved from data preparation to analysis.

APPLIED AI & MACHINE LEARNING PROJECTS

Clinical Trial Intelligence System | *Python, LangGraph, PubMedBERT, Cross-Encoder, FAISS, Streamlit, Git, Groq (gpt-oss-120b; built and evaluated on Llama 3.3 70B)* **May 2026**

github.com/Shrikant-Sharma/clinical-trial-rag | shrikant-clinical-rag.streamlit.app

- Designed and deployed an agentic Corrective RAG system on LangGraph over 484 ClinicalTrials.gov protocols (3,264 chunks): multi-step workflow with LLM-as-judge grading, query rewriting, conditional routing, and bounded retries. Three guardrails in layers reject adversarial queries scoring 0.91 cosine similarity that a single confidence threshold missed.
- Built the evaluation method: a six-question test set (must-answer, must-refuse and adversarial cases), RAGAS-style faithfulness and relevance scored by an LLM judge on a written rubric, and hand-run checks at two design decisions; the method showed the reranker cutting in-corpus refusals from 4 of 6 to 2 of 6, the remaining 2 being correct refusals.
- Engineered the retrieval stack from measured failure: compared 3 chunk sizes, switched to domain-tuned PubMedBERT embeddings after a general-purpose embedding scored a drug's brand and generic names as nearly unrelated, and added a cross-encoder reranker for 2-stage retrieval, cutting in-corpus refusals from 4 of 6 to 2 of 6. Deployed live on Streamlit with graceful rate-limit degradation.

Patient Risk Stratification Pipeline | *Python (Scikit-learn, XGBoost, PyTorch, SHAP, DoWhy, lifelines, scikit-survival), MLflow, FastAPI, Docker, AWS (Lambda, ECR, Mangum), Git* **May 2026**

github.com/Shrikant-Sharma/patient-risk-stratification | Live API on AWS Lambda

- Delivered an end-to-end ML pipeline for clinical heart disease risk stratification on UCI Heart Disease (303 patients, 13 clinical features); compared 5 model families with MLflow experiment tracking and 5-fold cross-validation (test AUCs within 2 points, 0.95 to 0.97) and selected tuned XGBoost for exact SHAP attributions and calibration.
- Added SHAP TreeExplainer for global and per-patient feature attributions critical for clinical decision support, and validated probability calibration at Brier 0.092 with reliability curves (Platt scaling rejected at 0.104).
- Extended the pipeline to causal inference and survival analysis on the NHEFS cohort (1,629 subjects); estimated treatment effects with DoWhy (propensity score matching and G-computation agreeing within 0.2 kg of the published reference) and modeled 10-year mortality with Cox PH and Random Survival Forest, showing that an unadjusted log-rank difference (p=0.005) disappeared after adjusting for age and other factors.
- Deployed live as a public REST API on AWS Lambda via Amazon ECR and Mangum: single-stage Docker image, EventBridge warmer keeping warm calls under a second, FastAPI docs, and MLflow experiment tracking.

Payment Transaction Anomaly Detection | *Python (Pandas, Scikit-learn, PySpark), Snowflake, Tableau, Git* **May 2026**

github.com/Shrikant-Sharma/pharma-compliance-spend-analytics | public.tableau.com/app/profile/shrikant.sharma

- Developed a payment-transaction anomaly detection system on the public Open Payments data (~15M general payments): a 10% sample of physician payments (989K rows) rolled up by provider ID to 289K physicians, with 5 entity-level features.
- Applied within-segment z-scores plus global IQR with a monetary floor and concentration logic, producing a ranked review queue; with no ground truth, results are reported as flag rates, not precision.
- Layered Isolation Forest anomaly detection on top; reconciliation against the rule-based flags, with Cohen's kappa 0.29 between the two methods, surfaced 3 distinct anomaly archetypes (caught by BOTH methods, rules-only, and model-only), revealing suspicious patterns no single method catches alone.
- Built the feature step in Pandas and PySpark with row-by-row matches on all five features; Snowflake SQL and 2 interactive Tableau dashboards surfaced payment concentration among the top counterparties.

EDUCATION

Clark University, Worcester, MA | Master of Science in Data Analytics **May 2022**

Sant Gadge Baba Amravati University, India | B.E. in Electronics & Telecommunication **Feb 2016**

ACHIEVEMENTS

Acclaim Award (3-Star) | Amgen | For data science delivery and cross-functional collaboration **Feb 2025**